Special wffs

August 19, 2025

We take a look at some special classes of wffs, that will play an important role in PL.

Definition 0.1. (**Tautology**)
α is a *tautology* if and only if under every valuation v, $v(\alpha) = T$.

Example: $p \vee \neg p$, or, $p \to p$.

Definition 0.2. (**Contradiction**)
α is a *contradiction* if and only if under every valuation v, $v(\alpha) = F$.

Examples: $p \wedge \neg p$, $\neg(p \to (q \to p))$.

Definition 0.3. (**Logically equivalent wffs**)
Two wffs α,β are said to be *logically equivalent* (written as $\alpha \equiv \beta$), if and only if $\alpha \leftrightarrow \beta$ is a tautology. Equivalently (*prove!*), $\alpha \equiv \beta$, provided under every valuation v, $v(\alpha) = v(\beta)$.

Example: $p \wedge q \equiv q \wedge p$; $p \vee q \equiv q \vee p$.

- Observe that wffs with the same propositional variables are logically equivalent, if and only if they have the same truth tables.